Dagger Categories of Relations

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Joint work with Chris Heunen,
Jean-Simon Pacaud Lemay, Paolo Perrone and Dario Stein

LOCAL SEMINAR
OCTOBER 2026

Bijection between contractions and relations in category of coisometries

Linear maps $f : X \to Y$ between Hilbert spaces such that $\|fx\| \leqslant \|x\|$ for all $x \in X$

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Isomorphism classes of jointly monic spans

Adjointable maps $f : X \to Y$ such that $ff^\dagger = 1$ (orthogonal projections onto closed subspaces)

But category of coisometries

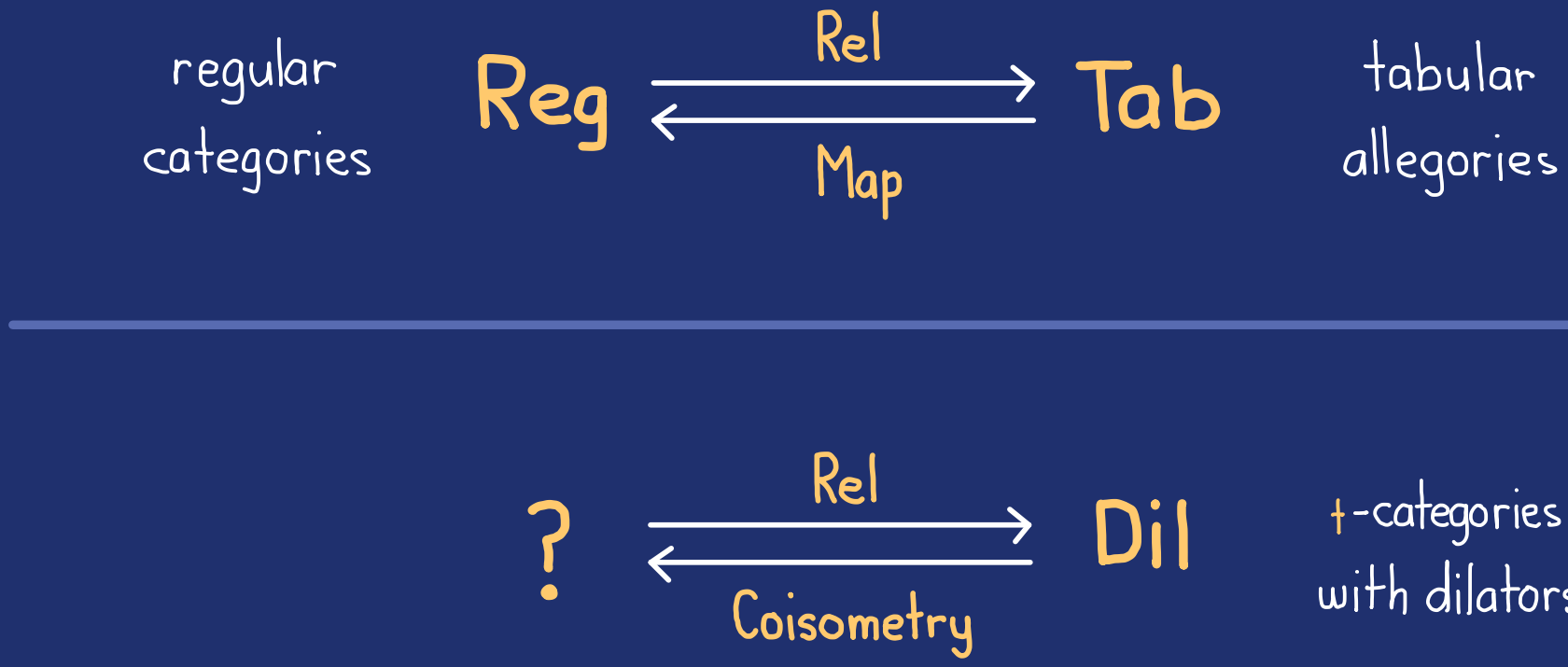
is not regular

But category of coisometries

is not regular

It does not have all pullbacks


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Dil
†-categories
with dilators

DEFINITION

A †-category is a category with $f^\dagger : Y \to X$ for each $f : X \to Y$ such that

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Hilbert spaces and contractions

$$\langle y | fx \rangle = \langle f^\dagger y | x \rangle$$

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The Bayesian inverse $f^\dagger$ of f is given by

$$\mathbb{P}_f(y|x)\, \mathbb{P}_X(x) = \mathbb{P}_{f^\dagger}(x|y)\, \mathbb{P}_Y(y)$$

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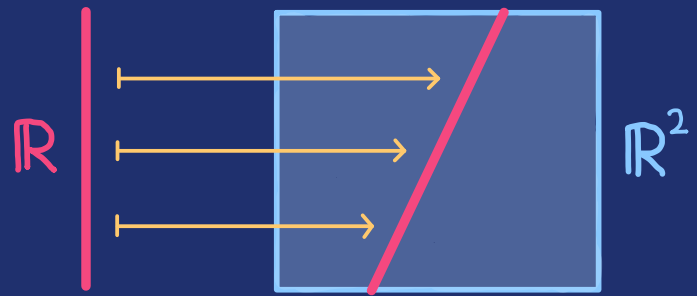
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Isometries in **Con** are inclusions of closed subspaces.



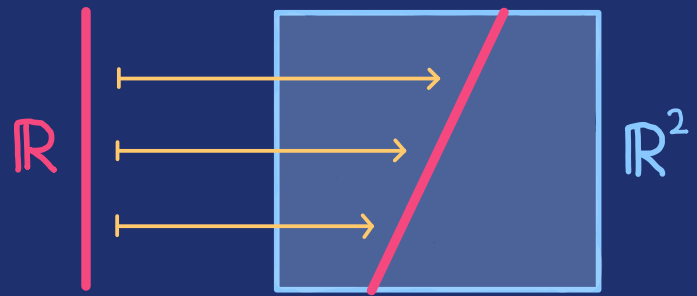
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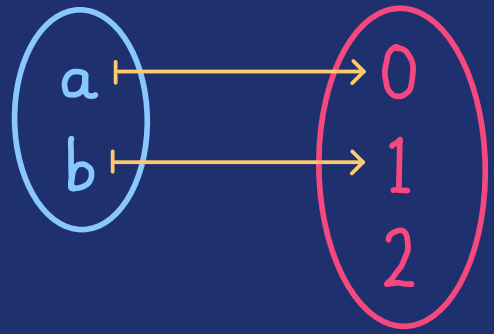
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Isometries in **PInj** are total.



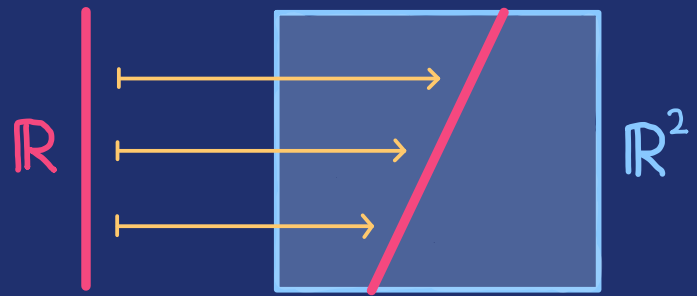
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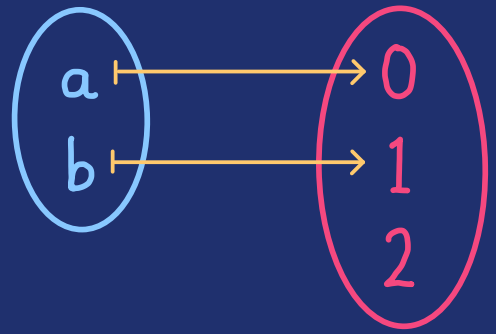
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Coisometries in **FinPS** are deterministic

$$\delta_X : (\Omega, \mathbb{P}) \to (A, X_* \mathbb{P})$$

$$\mathbb{P}_{\delta_X}(a \mid \omega) = \begin{cases} 1 & \text{if } a = X(\omega) \\ 0 & \text{otherwise} \end{cases}$$

random variable $X : \Omega \to A$.

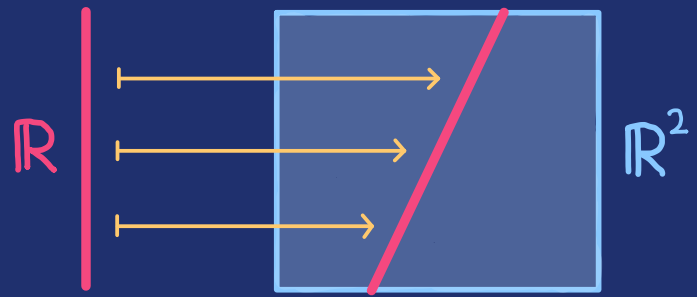
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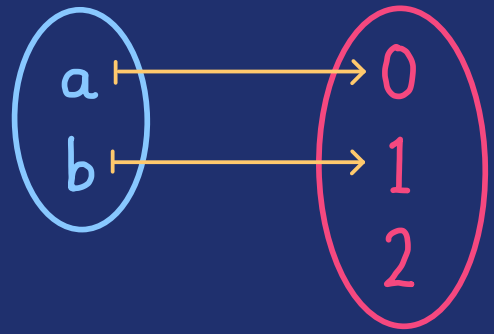
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$$\delta_Y \delta_X^\dagger(b \mid a) = \mathbb{P}[Y = b \mid X = a]$$

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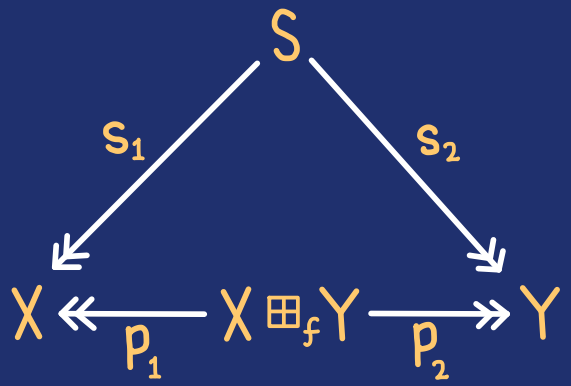
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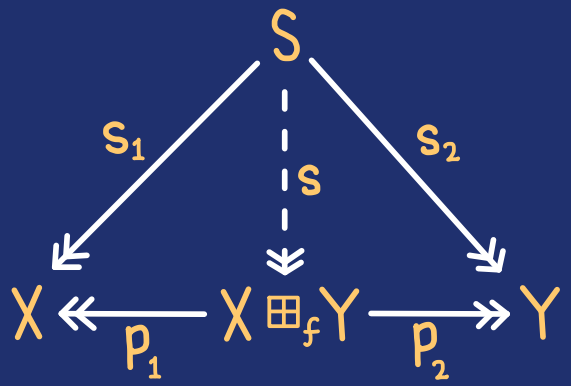
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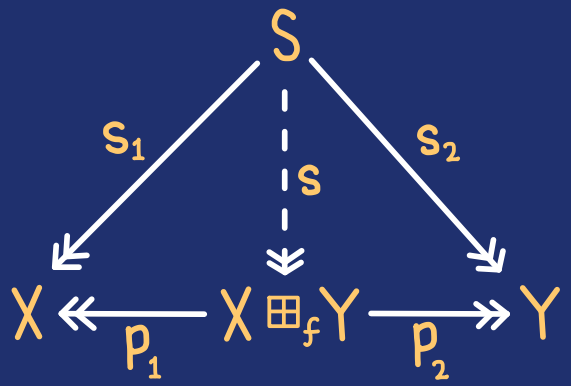
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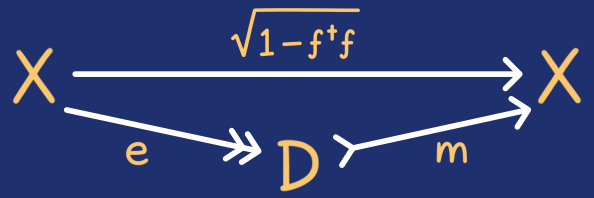
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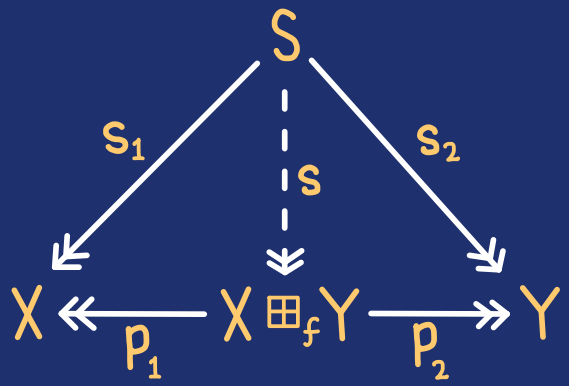
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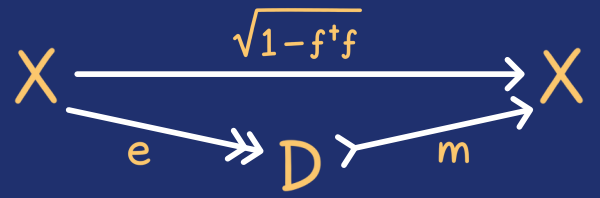
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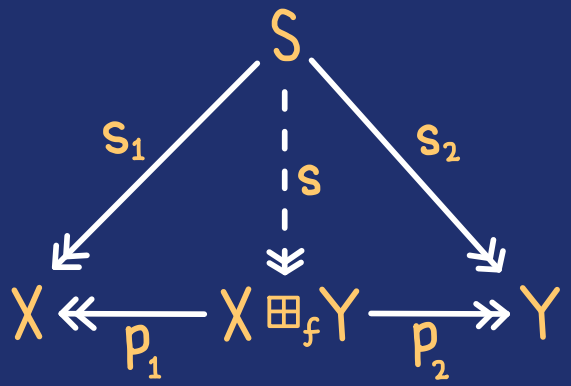
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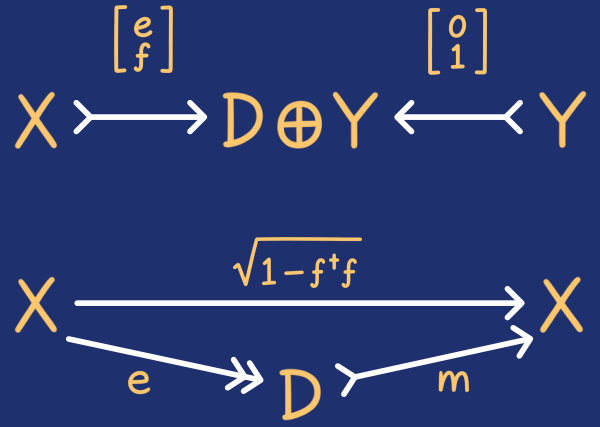
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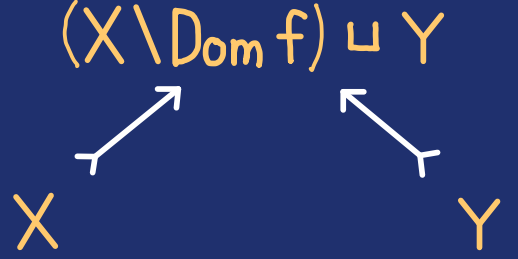
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Codilators in **PInj**



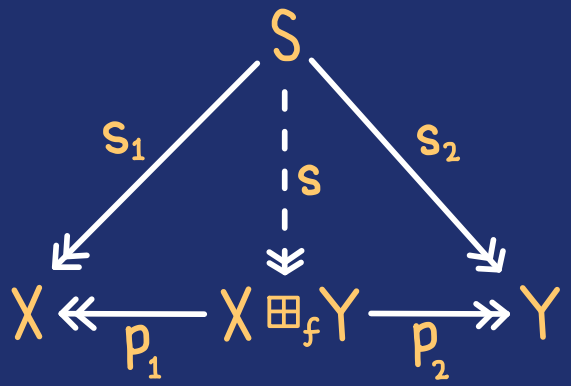
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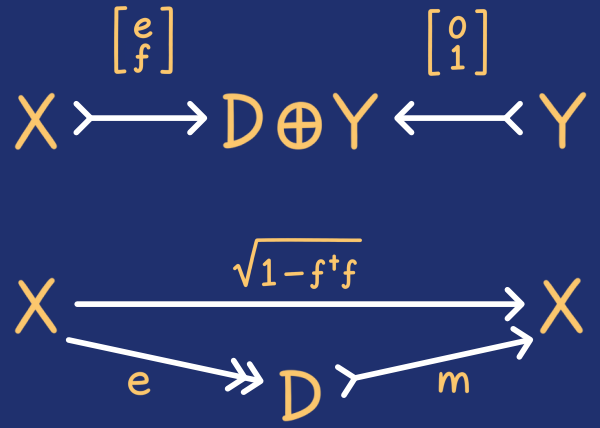
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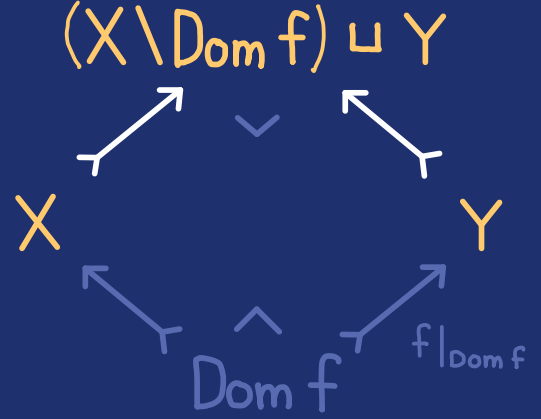
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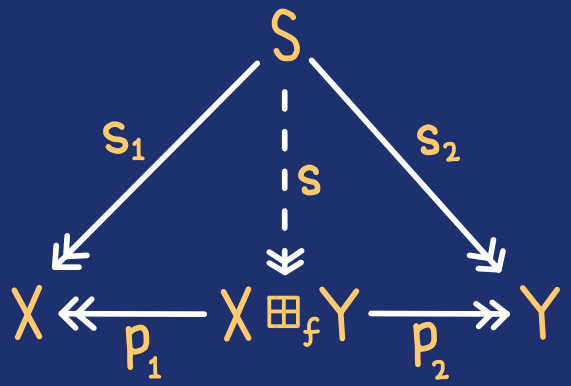
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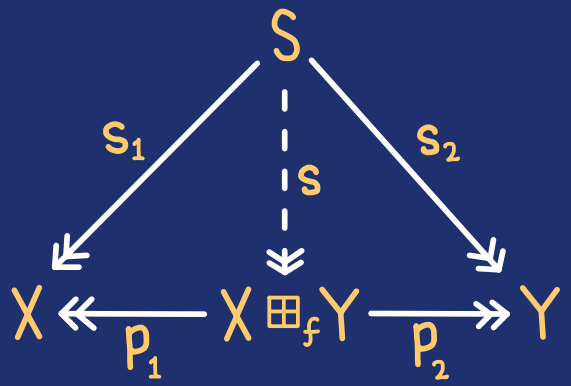
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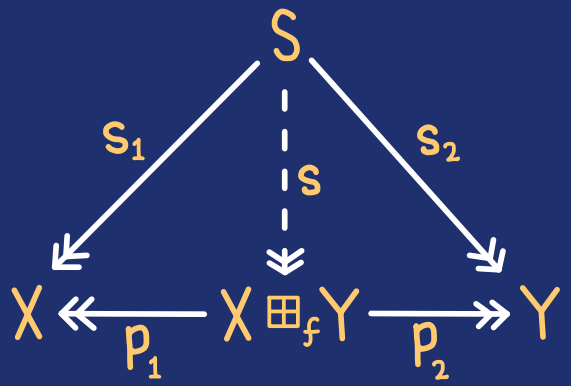
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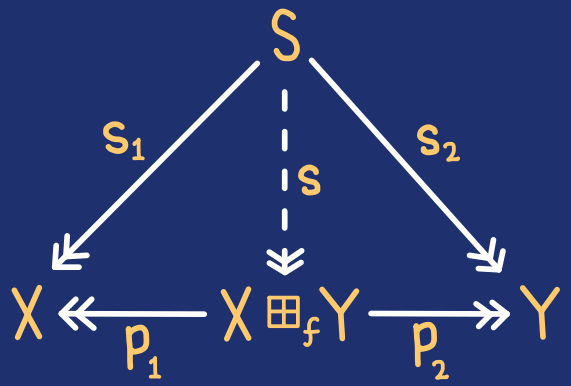
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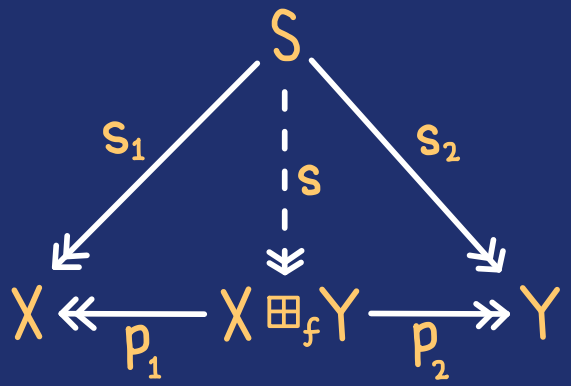
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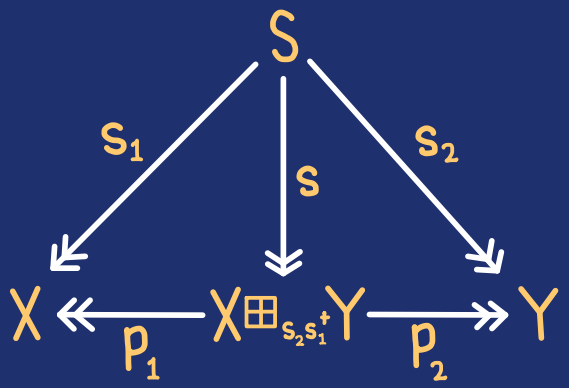
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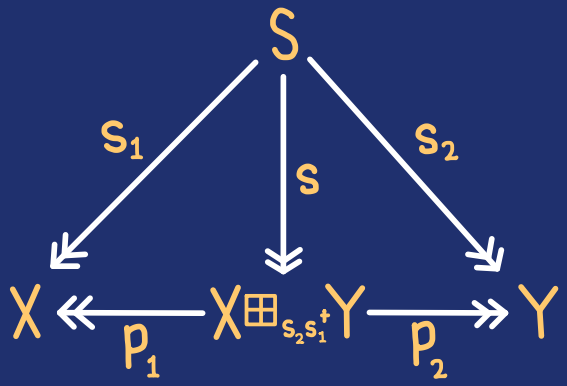
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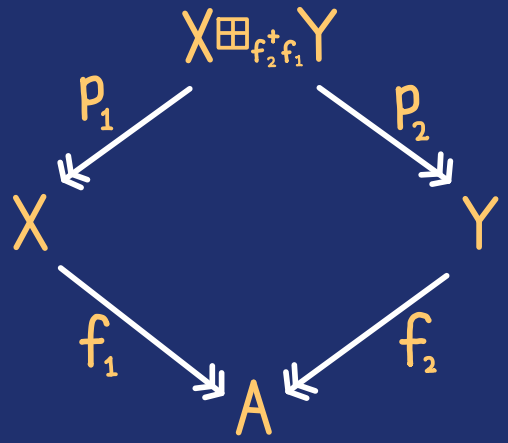


factorisation of spans

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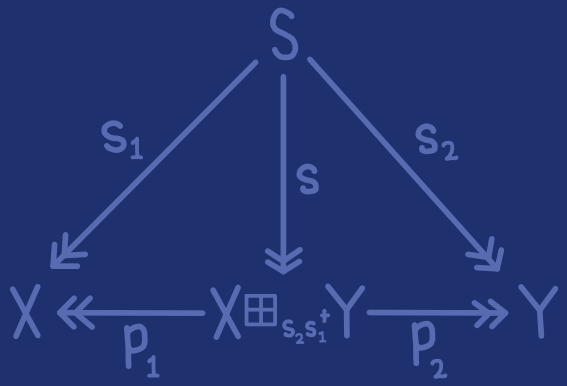


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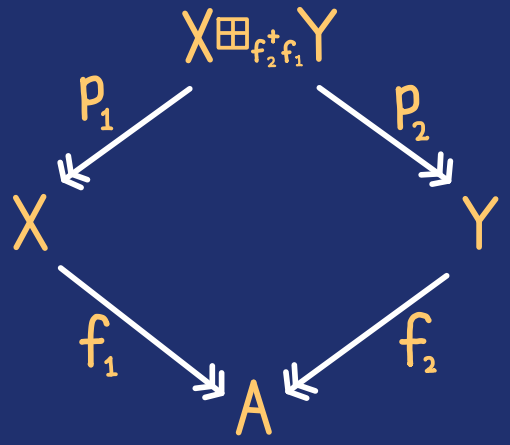


"pullbacks"

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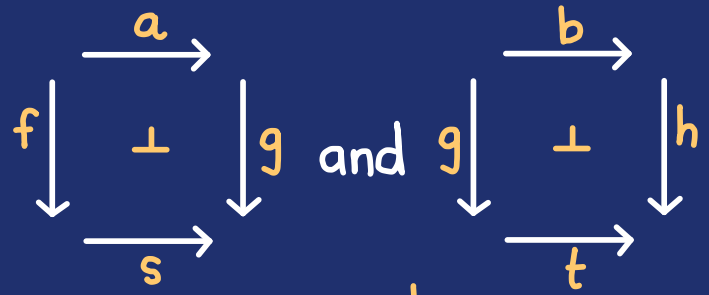
An independence category is a category equipped with a predicate $\perp$ on commuting squares such that

Simpson, "Equivalence and conditional independence in atomic sheaf logic"

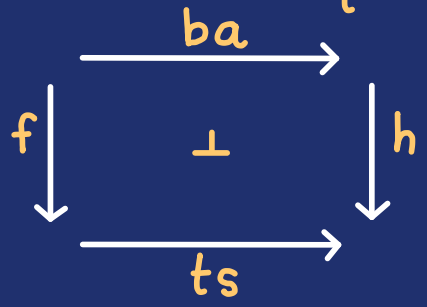
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An independence category is a category equipped with a predicate $\perp$ on commuting squares such that

(i)



implies



Simpson, "Equivalence and conditional independence in atomic sheaf logic"

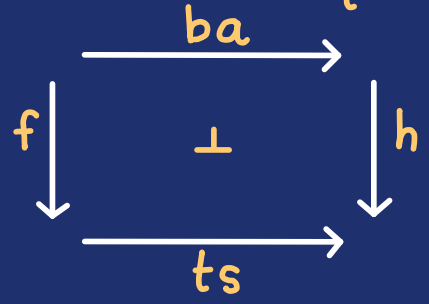
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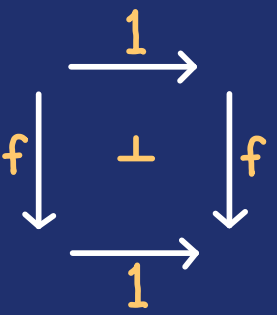
(i)



implies



(ii)



Simpson, "Equivalence and conditional independence in atomic sheaf logic"

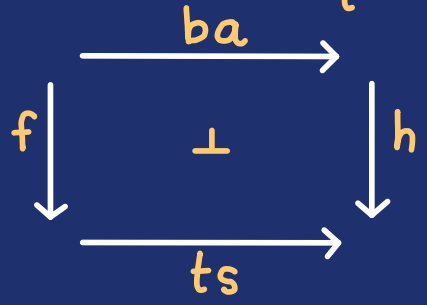
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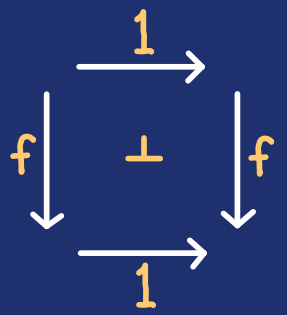
(i)



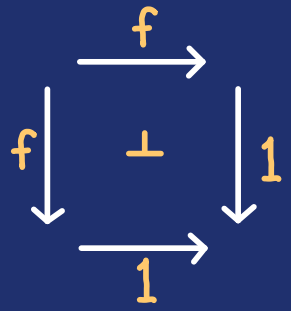
implies



(ii)



(iii)



Simpson, "Equivalence and conditional independence in atomic sheaf logic"

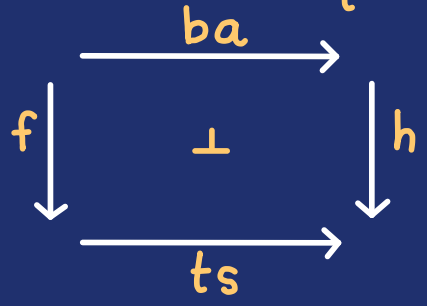
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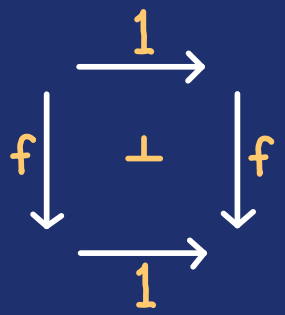
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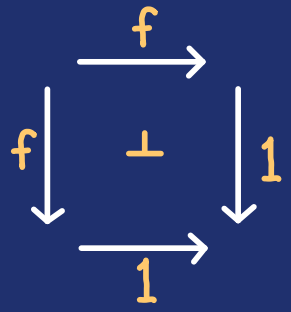
implies



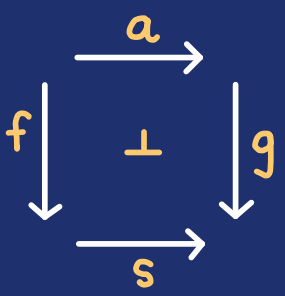
(ii)



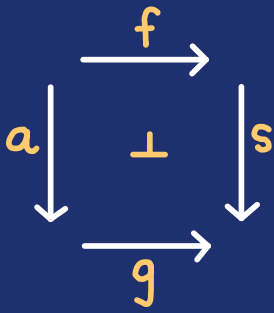
(iii)



(iv)



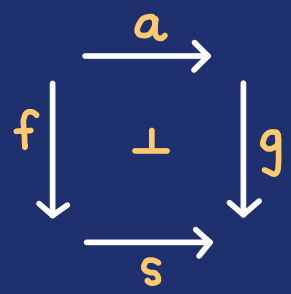
implies



Simpson, "Equivalence and conditional independence in atomic sheaf logic"

LEMMA

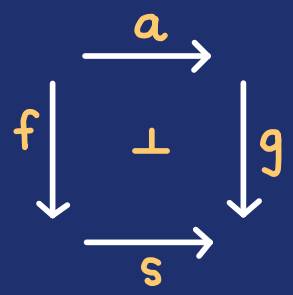
If $\mathbf{C}$ is a $\dagger$-category, then $\mathbf{Coisometry}(\mathbf{C})$ is an independence category where



if $ga = sf$ and $af^\dagger = g^\dagger s$

LEMMA

If $\mathbf{C}$ is a $\dagger$-category, then $\mathbf{Coisometry}(\mathbf{C})$ is an independence category where

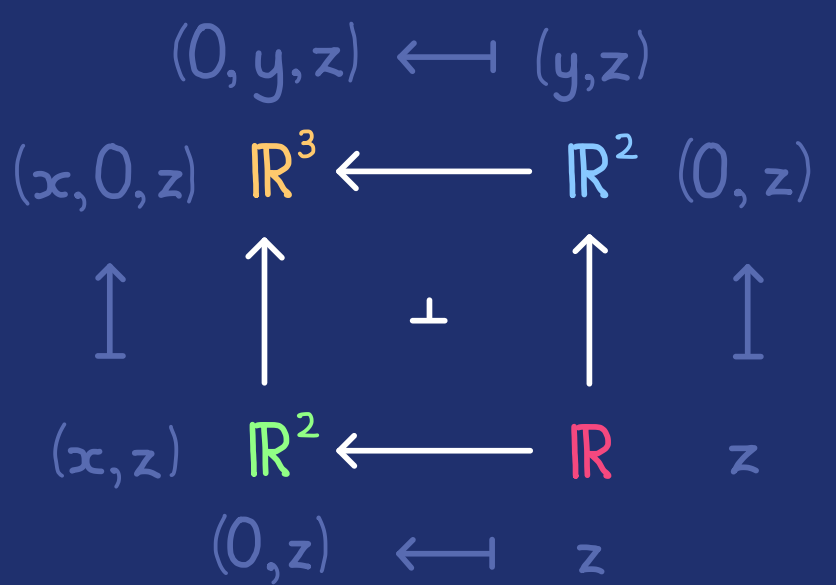


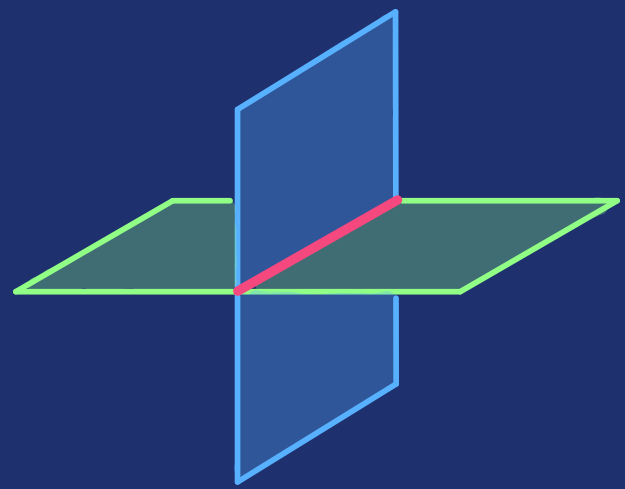
if $ga = sf$ and $af^\dagger = g^\dagger s$

EXAMPLES

In $\mathbf{Isometry}(\mathbf{Con})$,

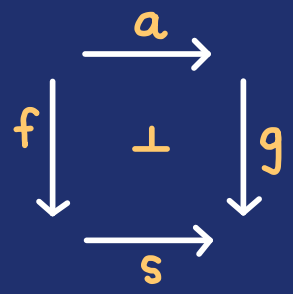
$\perp$ captures relative orthogonality.





LEMMA

If $\mathbf{C}$ is a $\dagger$-category, then $\mathbf{Coisometry}(\mathbf{C})$ is an independence category where

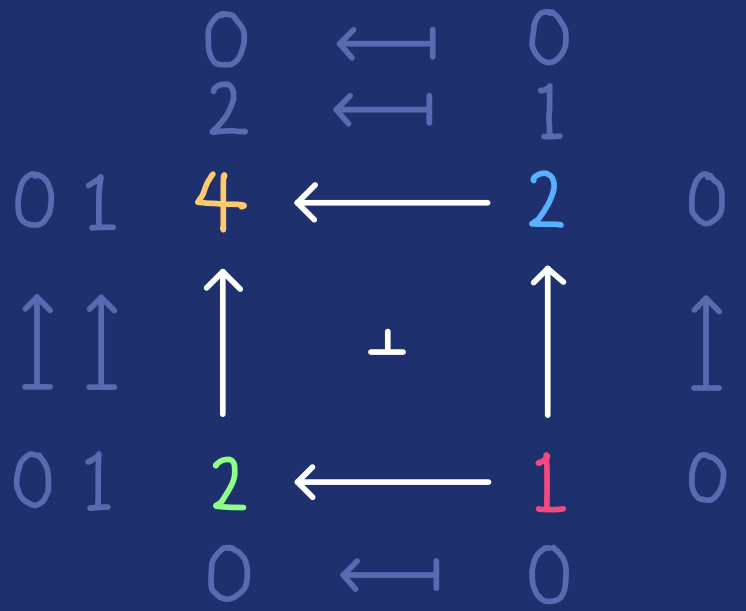


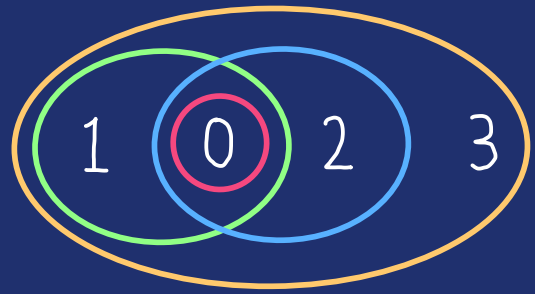
if $ga = sf$ and $af^{\dagger} = g^{\dagger}s$

EXAMPLES

In $\mathbf{Isometry}(\mathbf{PInj})$,

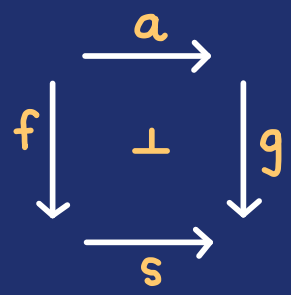
$\perp$ captures relative disjointness.





LEMMA

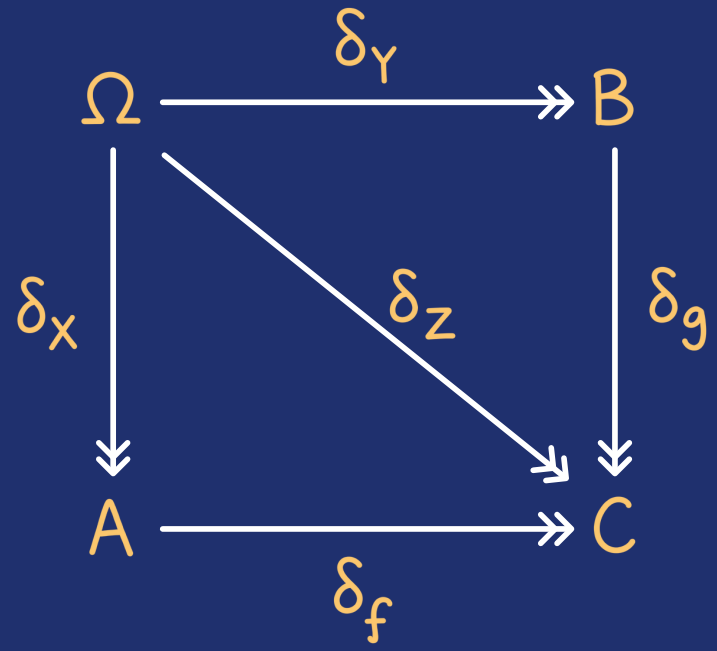
If $\mathbf{C}$ is a $\dagger$-category, then $\mathbf{Coisometry(C)}$ is an independence category where



if $ga = sf$ and $af^\dagger = g^\dagger s$

EXAMPLES

In $\mathbf{Coisometry(FinPS)}$,



is independent iff

$$\mathbb{P}[Y = b \mid X = a] = \mathbb{P}[Y = b \mid Z = fa]$$

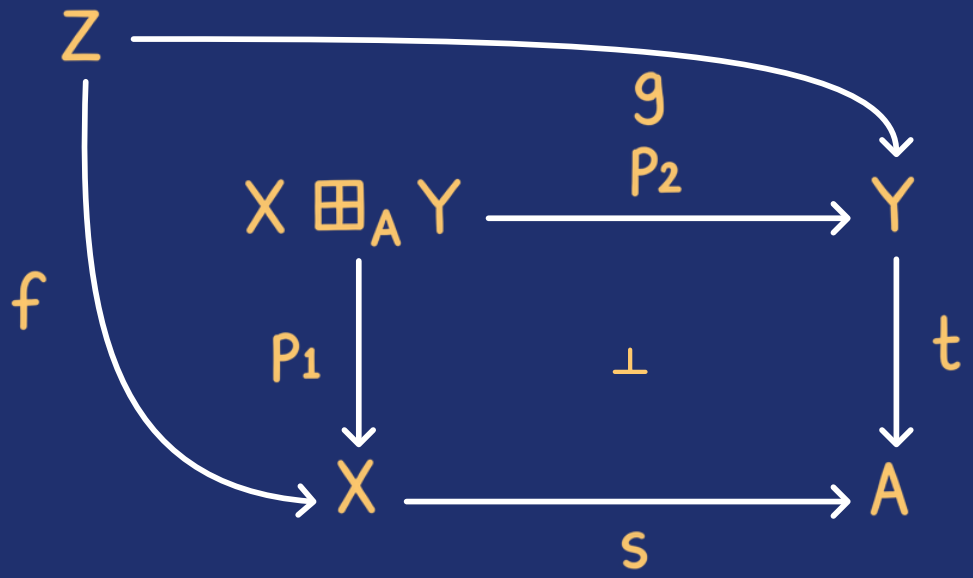
DEFINITION

independent pullback



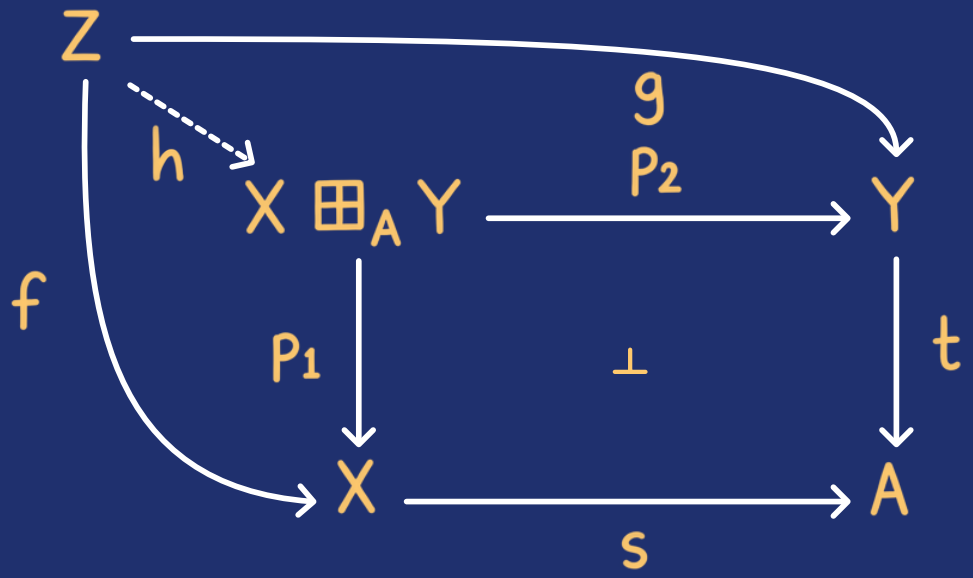
DEFINITION

independent pullback



DEFINITION

independent pullback

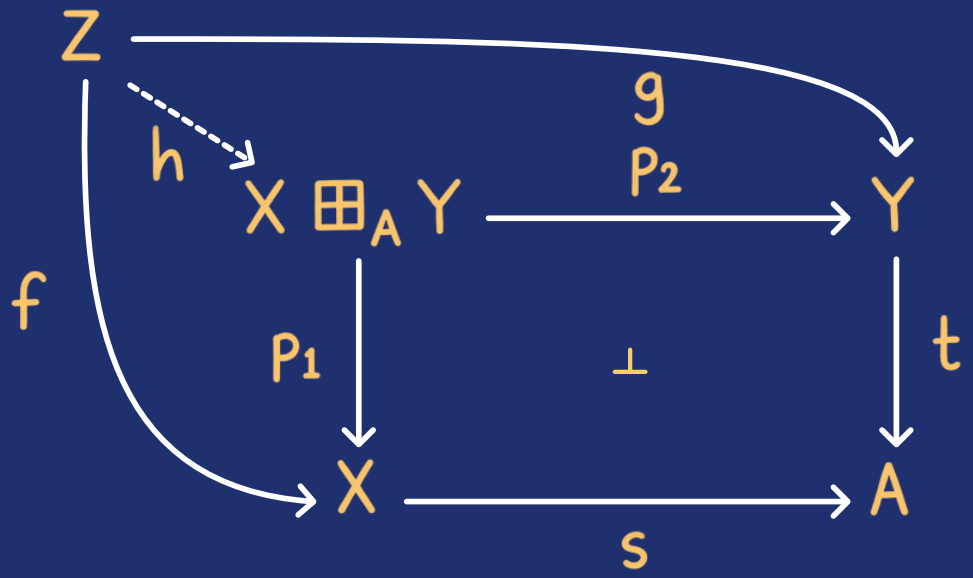


LEMMA

If C is a †-category with dilators, then Coisometry(C) has independent pullbacks.

DEFINITION

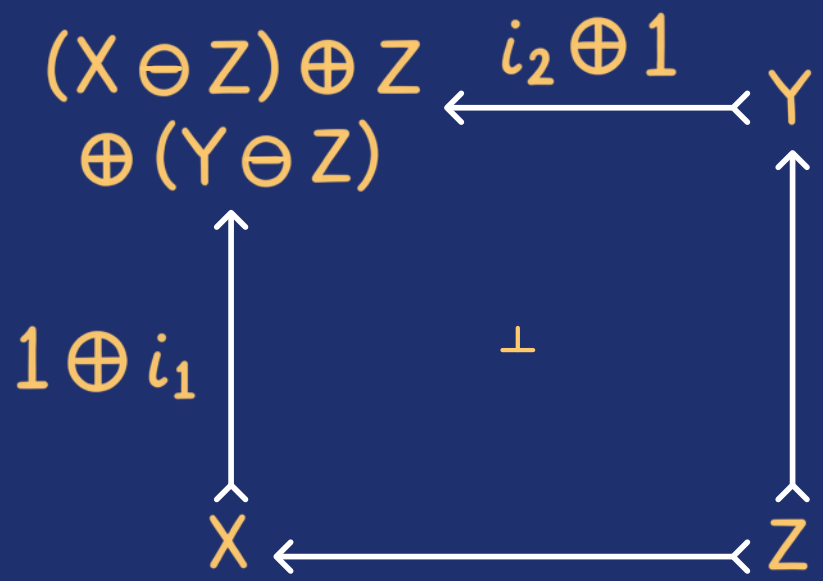
independent pullback



LEMMA

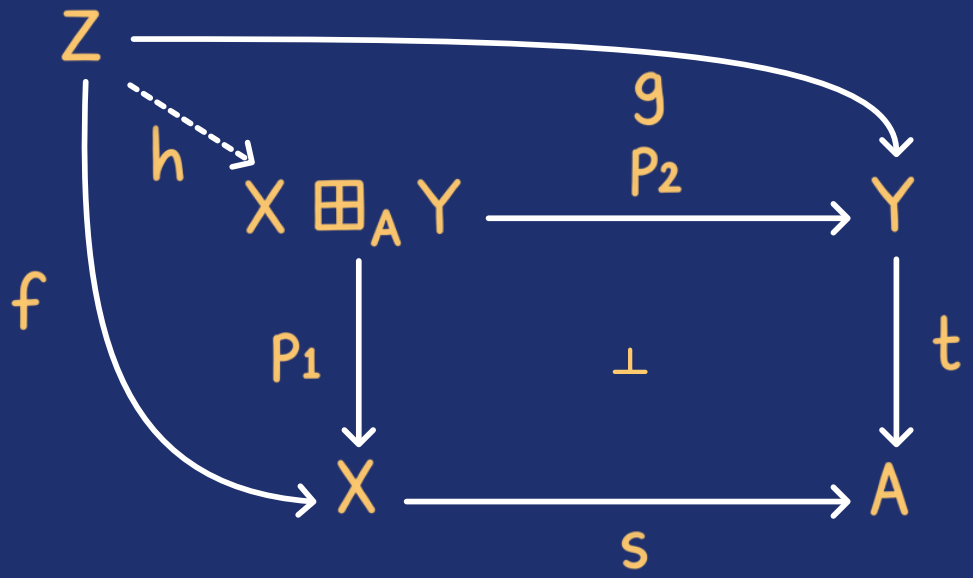
If C is a †-category with dilators, then Coisometry(C) has independent pullbacks.

Con



DEFINITION

independent pullback

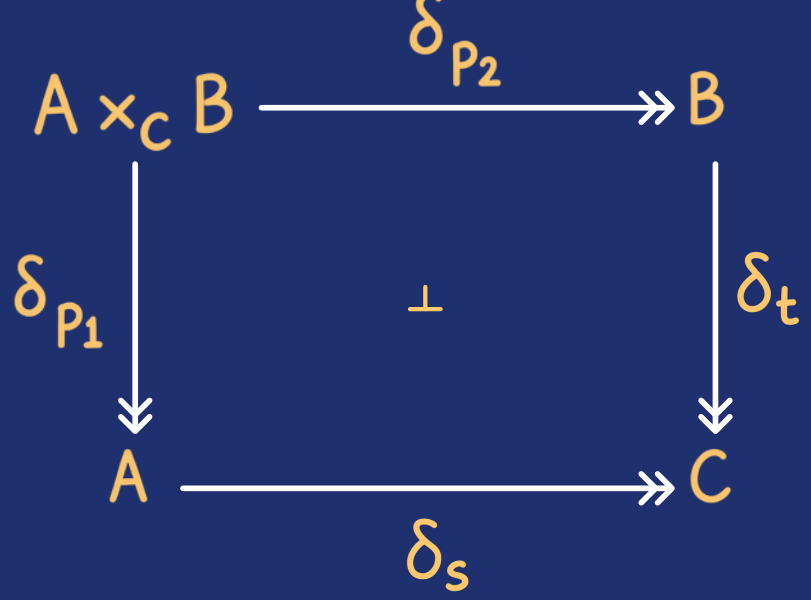


LEMMA

If C is a t-category with dilators, then Coisometry(C) has independent pullbacks.

FinPS

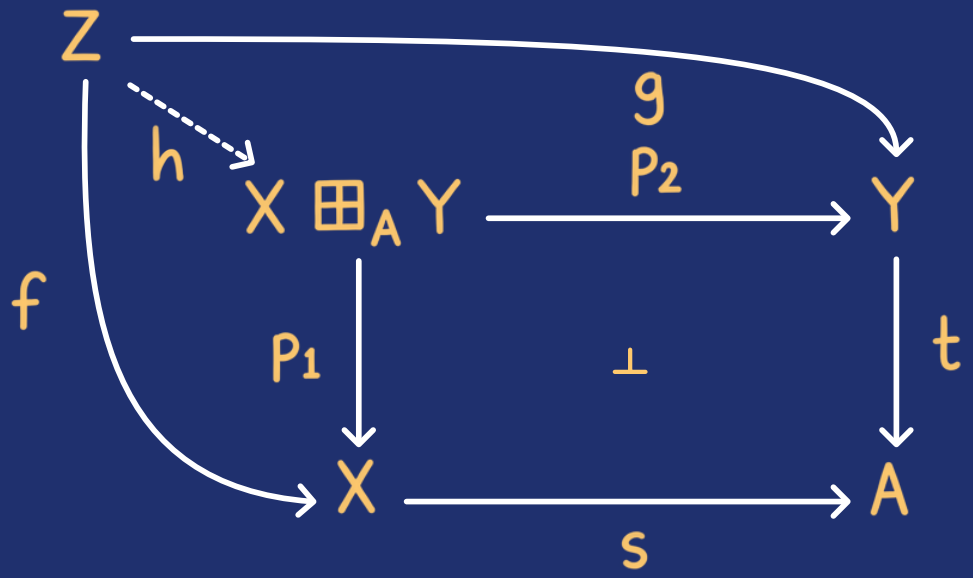
conditional product



$$\mathbb{P}_{A \times_C B}(a,b) = \frac{\mathbb{P}_A(a)\,\mathbb{P}_B(b)}{\mathbb{P}_C(sa)}$$

DEFINITION

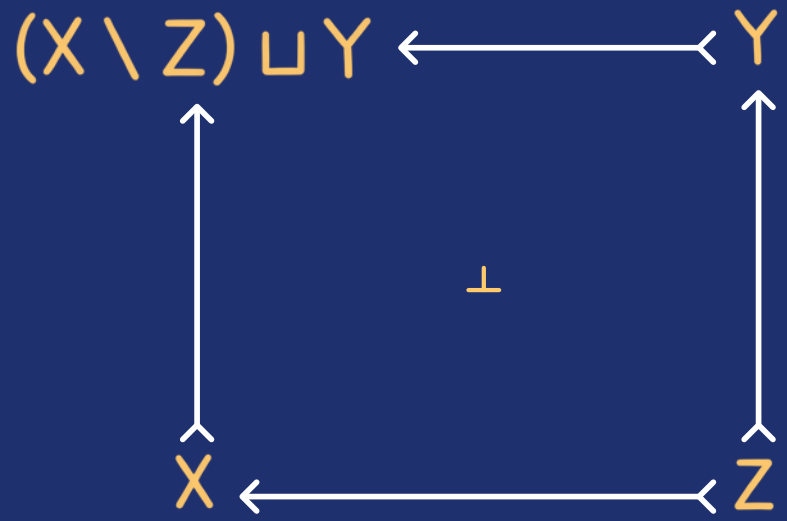
independent pullback



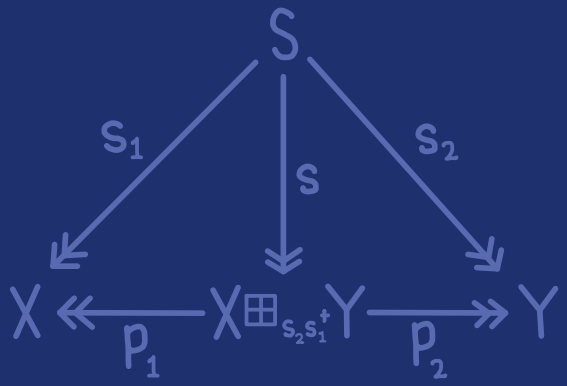
LEMMA

If C is a †-category with dilators, then Coisometry(C) has independent pullbacks.

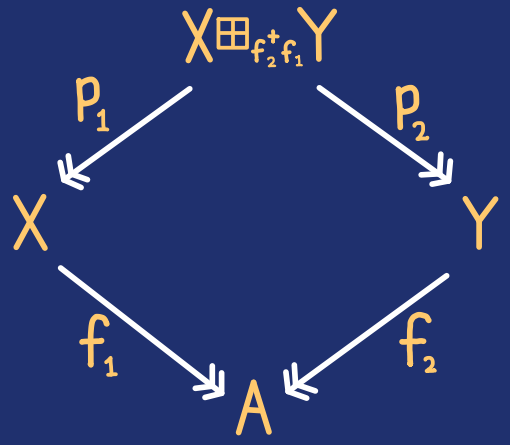
PInj



Remnants of dilators in Coisometry(C)

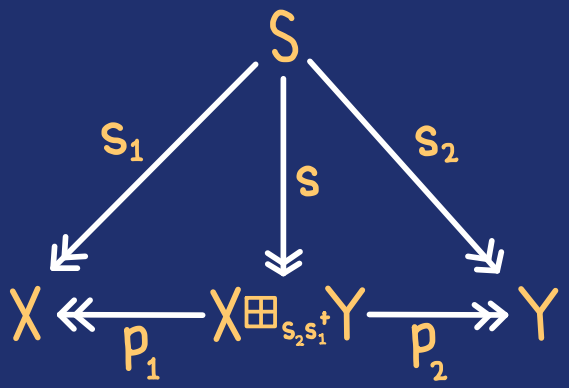


factorisation of spans

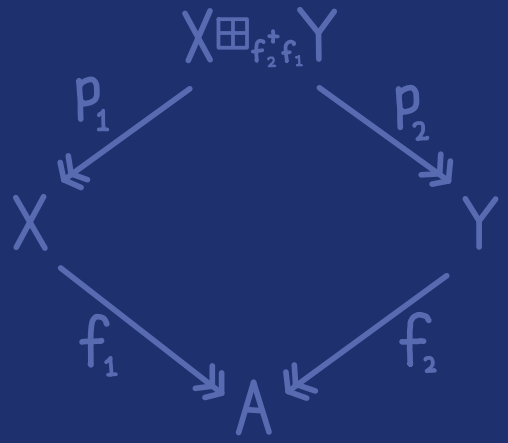


"pullbacks"

Remnants of dilators in Coisometry (C)



factorisation of spans



"pullbacks"

Let $\mathbb{C}$ be a $\dagger$-category with dilators.

LEMMA

A span of coisometries



is jointly monic iff it is a dilator.

Let $\mathbb{C}$ be a $\dagger$-category with dilators.

LEMMA

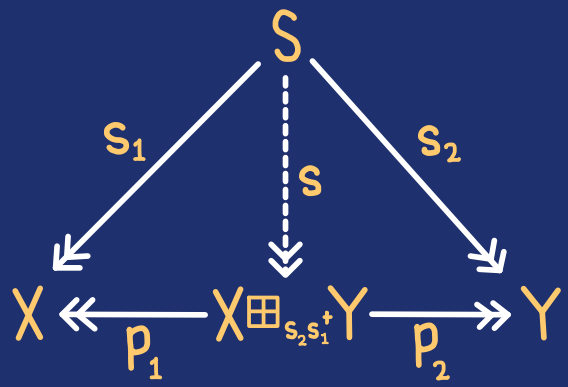
A span of coisometries

$$X \xleftarrow{s_1} S \xrightarrow{s_2} Y$$

is jointly monic iff it is a dilator.

COROLLARY

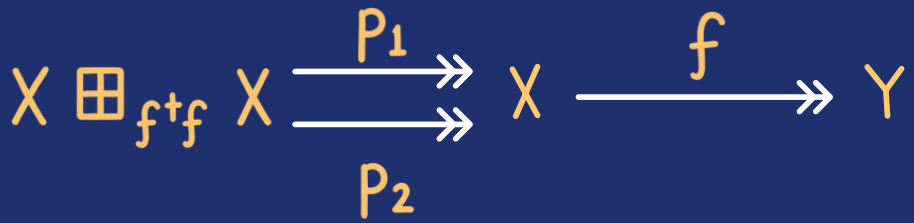
Every span of coisometries factors through a jointly monic span.



Let $\mathbf{C}$ be a $\dagger$-category with dilators.

LEMMA

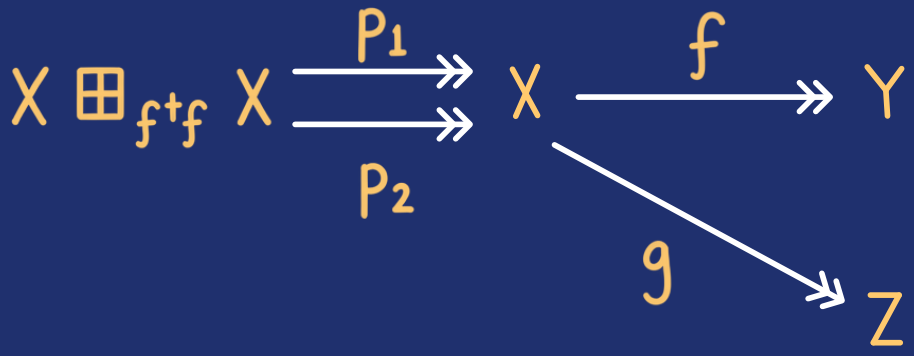
Every morphism in $\mathbf{Coisometry}(\mathbf{C})$ is the coequaliser of its independent kernel pair



Let $\mathbb{C}$ be a $\dagger$-category with dilators.

LEMMA

Every morphism in Coisometry($\mathbb{C}$) is the coequaliser of its independent kernel pair

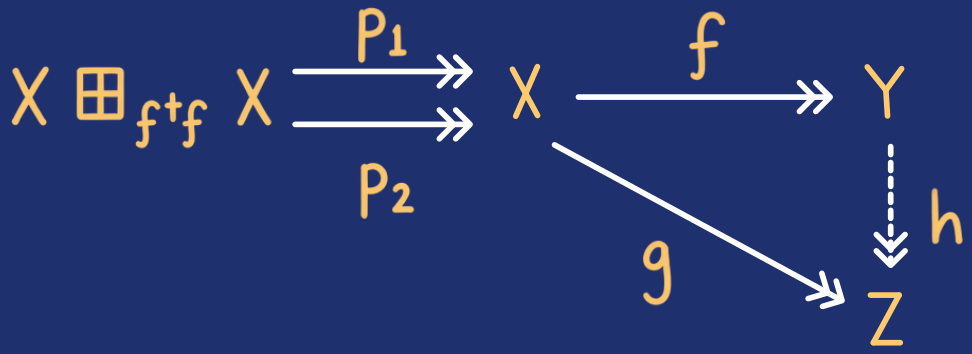


Suppose that $gP_1 = gP_2$

Let $\mathbf{C}$ be a $\dagger$-category with dilators.

LEMMA

Every morphism in $\mathbf{Coisometry}(\mathbf{C})$ is the coequaliser of its independent kernel pair



Suppose that $gp_1 = gp_2$

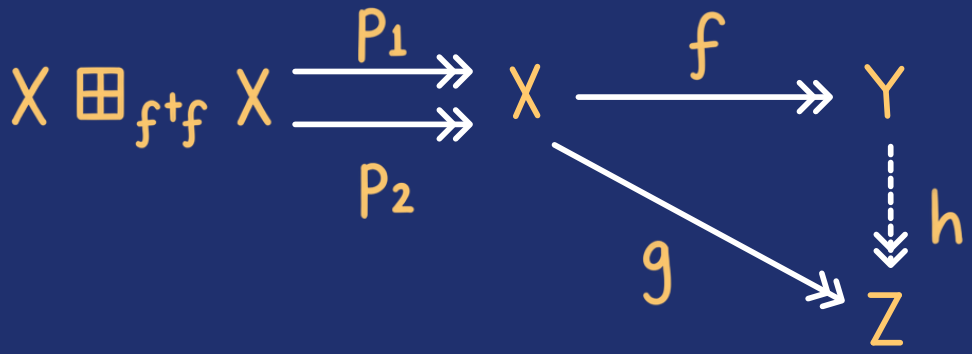
If h exists, then

$$h = hff^\dagger = gf^\dagger$$

Let $\mathbf{C}$ be a $\dagger$-category with dilators.

LEMMA

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Suppose that $gp_1 = gp_2$

If h exists, then

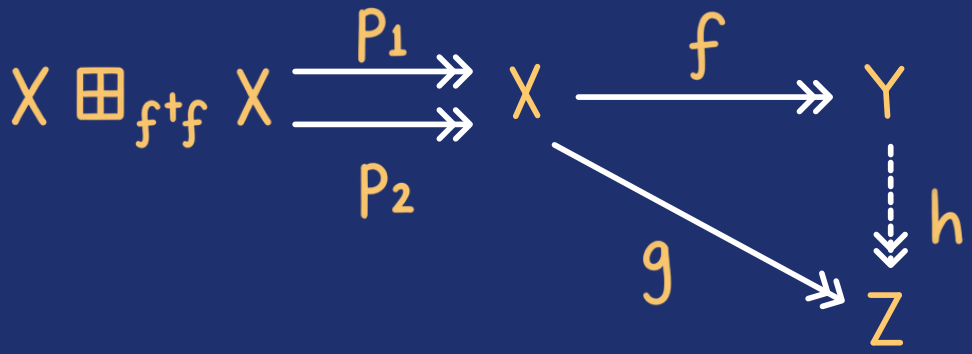
$$h = hff^\dagger = gf^\dagger$$

Letting $h = gf^\dagger$,

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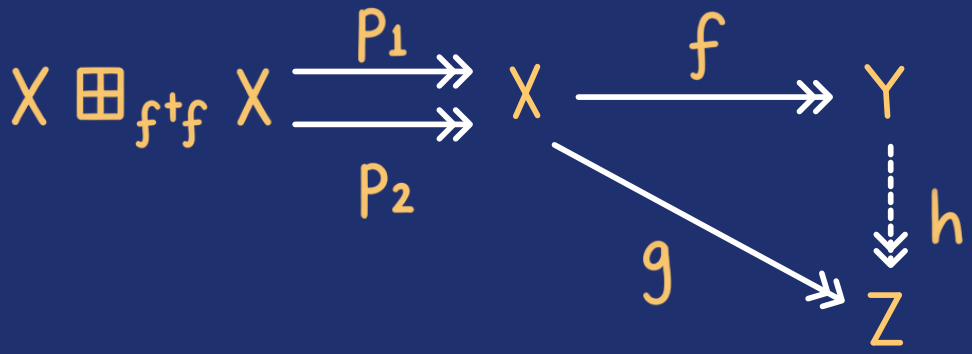
Letting $h = gf^\dagger$,

$$hf = gf^\dagger f = gp_2p_1^\dagger$$
$$= gp_1p_1^\dagger = g$$

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If h exists, then

$$h = hff^\dagger = gf^\dagger$$

Letting $h = gf^\dagger$,

$$hf = gf^\dagger f = gp_2p_1^\dagger$$
$$= gp_1p_1^\dagger = g$$

$$hh^\dagger = hff^\dagger h^\dagger = gg^\dagger = 1.$$

DEFINITION

A category is regular if

(*i*) every cospan has a pullback,

(*ii*) every span factors as a regular epic followed by a jointly monic span,

(*iii*) regular epics are pullback-stable.

DEFINITION

An independence category is epi-regular if

(*i*) every cospan has an independent pullback,

(*ii*) every span factors through a jointly monic span,

(*iii*) every morphism is regular epic.

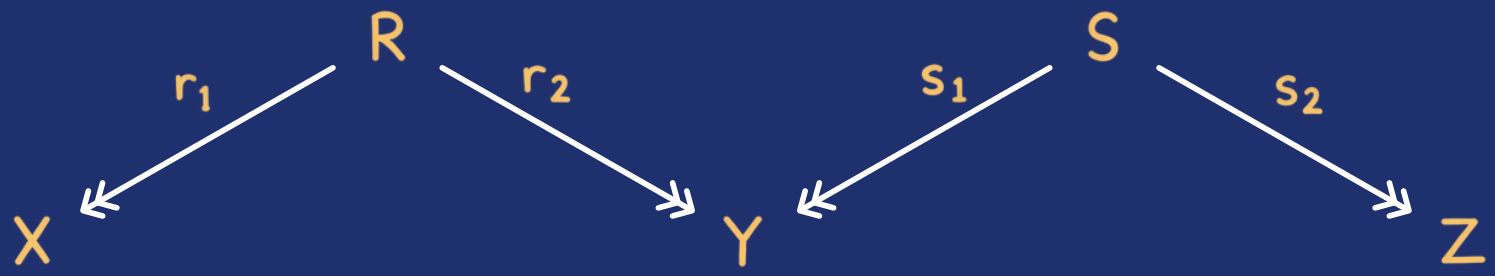
THEOREM

If $\mathbf{C}$ is a $\dagger$-category with dilators, then $\mathbf{Coisometry(C)}$ is epi-regular.

DEFINITION

The †-category Rel(D) of relations in an epi-regular independence category D has:

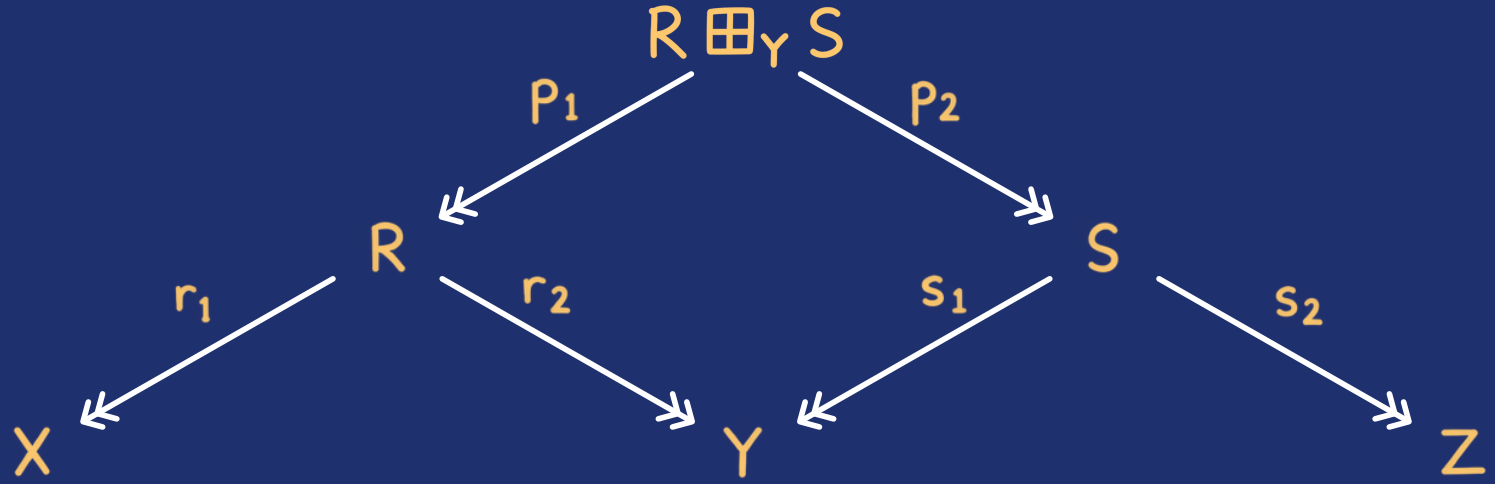
- objects the objects of D
- morphisms the isomorphism classes of jointly monic spans



DEFINITION

The †-category Rel(D) of relations in an epi-regular independence category D has:

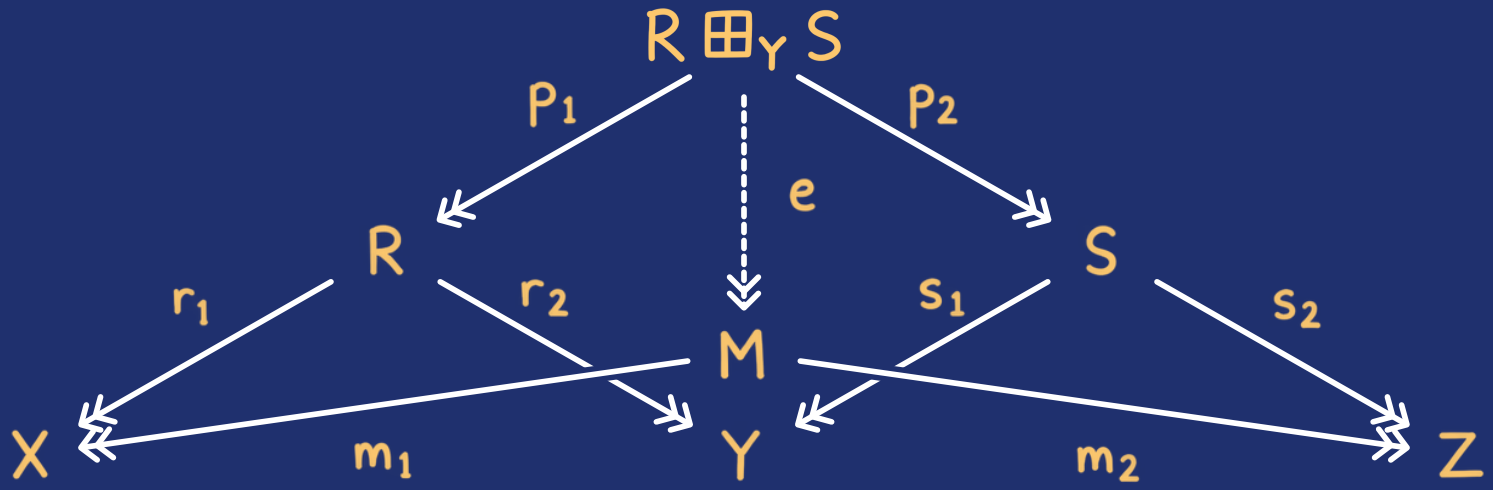
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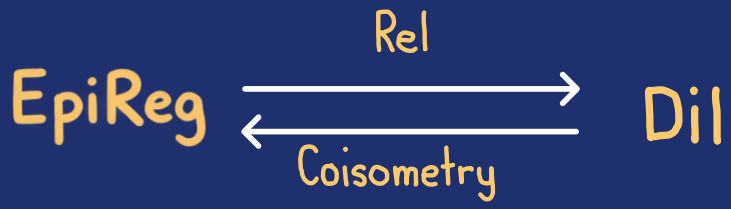
- objects the objects of D
- morphisms the isomorphism classes of jointly monic spans



PROPOSITION Rel(D) is a †-category with dilators.

THEOREM

Rel and **Coisometry** form an adjoint equivalence.



$D \cong \mathrm{Coisometry}(\mathrm{Rel}(D))$ $\qquad$ $\mathrm{Rel}(\mathrm{Coisometry}(C)) \cong C$

$f \mapsto [1, f]$ $\qquad$ $[s_1, s_2] \mapsto s_2 s_1^\dagger$

mattd@lmf.cnrs.fr

https://mdimeglio.github.io

DAGGER CATEGORIES OF RELATIONS

THE EQUIVALENCE OF DILATORY DAGGER CATEGORIES AND EPI-REGULAR INDEPENDENCE CATEGORIES

MATTHEW DI MEGLIO, CHRIS HEUNEN, JEAN-SIMON PACAUD LEMAY, PAOLO PERRONE, AND DARIO STEIN

Abstract. Several categories look like categories of relations, but do not fit the established theory of relations in regular categories. They include the category of surjective multivalued functions, the category of injective partial functions, the category of finite probability spaces and stochastic matrices, and the category of Hilbert spaces and linear contractions. To explain these anomalous examples, we develop a parallel theory of relations in *epi-regular independence categories*. Just as regular categories correspond to tabular allegories, epi-regular independence categories correspond to *dilatory dagger categories*. The equivalence maps epi-regular independence categories to their associated dagger category of relations, and dilatory dagger categories to their wide subcategory of coisometries.